ENTERPRISE DATA ANALYTICS & FULL-STACK PORTFOLIO

System Architecture & Project Blueprint

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Executive Overview

A production-grade, full-stack Data Analytics showcase engineered with React 18, TypeScript, Three.js, Node.js, and relational database persistence. The application features interactive 3D WebGL data visualizations, flagship Power BI dashboard case studies analyzing $2.4M+ in enterprise sales, a real-time dual contact communication suite with automated Gmail SMTP notification dispatch, and a secure decoupled admin management portal.

1. CORE PERFORMANCE & PROJECT IMPACT METRICS

Quantified accomplishments delivered within this platform

$2.4M+	50,000+	100% Free	< 1.2s
Sales Analyzed	**Records Handled**	**Gmail Automation**	**Initial Page Load**
Flagship Power BI Dashboard	ETL & SQL Optimization	Direct Nodemailer SMTP Relay	Optimized Edge CDN Delivery

2. PUBLIC ACCESS & ZERO-TRUST SECURITY MODEL

Strict separation of public client and private cloud resources

In strict accordance with modern security standards, this repository maintains an absolute zero-trust boundary:

- **Public Client Deployment:** The public user portal is hosted on high-performance Edge infrastructure and accessible at https://ssportfoliomines.vercel.app/. No backend database connection strings or internal service credentials are ever exposed.
- **Isolated Cloud Backend:** The relational backend and database operate in an isolated cloud environment. Access is gated by strict CORS allowlists, IP restrictions, and rate limiters.
- **Credential Obfuscation:** All API keys, JWT secrets, and database passwords reside exclusively in protected cloud environment variables. Tracked repository files include only sanitized .env.example templates.
- **Automated Contact Relay:** Form submissions trigger server-side sanitized validation, transactional database insertion, and instant Nodemailer SMTP dispatch to the administrator without exposing mail server credentials.

3. SYSTEM ARCHITECTURE & COMPONENT INTERACTION FLOW

End-to-end request lifecycle and automated event pipelines

The diagram below illustrates the end-to-end data lifecycle across the public visitor interface, client-side rendering engines, cloud API microservices, relational persistence layers, and external notification gateways.

VISITOR & CLIENT LAYER (Edge CDN / Vercel)

Interactive 3D WebGL Canvas | Audio Engine | Glassmorphic Responsive UI | Live Capstone Gallery

COMMUNICATION & API GATEWAY LAYER (Node.js & Express)

Strict CORS * Input Validation * JWT Authentication * Rate Limiting * Real-Time SSE Event Stream

DATA PERSISTENCE LAYER (Relational MySQL Database)

Connection Pooling * Automated SQL Schema Migrations * Normalized Relational Tables * Offline Fallback

NOTIFICATION & EXTERNAL AUTOMATION GATEWAYS

1. Nodemailer SMTP: Instant direct alert to sirroshan999@gmail.com
2. Webhook Forwarding: Optional Make.com / Zapier trigger payloads
3. Instant WhatsApp & Phone Actions: Client-side dynamic protocol deep-linking

4. CONTACT FORM AUTOMATED PIPELINE (NODEMAILER)

Zero-cost, unlimited instant Gmail notification engine

When a recruiter or prospective collaborator submits an inquiry on the portfolio, the backend triggers a synchronized dual-path workflow:

Step 1: Path A - Transactional Storage: Sanitized payloads (Name, Email, Subject, Company, Message, IP, Timestamp) are saved to MySQL.

Step 2: Path B - Real-Time Admin Alert: Server-Sent Events (SSE) immediately broadcast a notification to the Netlify Admin Inbox.

Step 3: Path C - Direct Gmail Forwarding: Nodemailer establishes an encrypted TLS SMTP handshake and sends a formatted HTML email directly to the administrator's Gmail inbox, bypassing third-party subscription charges.

Step 4: Path D - Optional Webhook Relay: If configured, a JSON payload is dispatched to Make.com/Zapier for CRM integration.

5. RELATIONAL DATABASE SCHEMA & ENTITY-RELATIONSHIP (ERD)

Normalized structure ensuring data integrity and fast retrieval

The database architecture utilizes normalized MySQL schemas with indexed foreign keys, relational join tables, and transactional audit fields. Below is the specification of core entities:

PROFILES

Stores core identity, designation, executive bio, social URLs, and live KPI stats.

`Fields: id (PK), name, designation, tagline, about_bio, email, phone, location, linkedin_url, github_url, stats_json`

PROJECTS

Contains portfolio projects, difficulty ratings, live demo links, and comprehensive case studies.

`Fields: id (PK), title, project_type, short_description, category, difficulty, featured, case_study_json, date_created`

SKILLS & PROJECT_SKILLS

Technical competency catalog linked via many-to-many relationship mapping projects to applied skills.

`Fields: id (PK), name, category, proficiency_level, icon_url, description, tools_used_json | project_id (FK), skill_id (FK)`

EDUCATION & ACHIEVEMENTS

Academic timeline (Presidency College BCA) alongside certified honors (IT Quiz 2nd Place, NCC Sergeant).

`Fields: id (PK), degree/title, institution/issuer, period/issue_date, grade_gpa, highlights_json, credential_url`

CONTACT_MESSAGES & REPLIES

Incoming visitor inquiries tracked through lifecycle states (unread, read, replied, archived) with audit logs.

`Fields: id (PK), name, email, subject, company, message, status, reply_content, ip_address, created_at, updated_at`

6. ENTITY-RELATIONSHIP CARDINALITY OVERVIEW

Key relational dependencies and foreign constraints

```
[PROFILES]  1  % % % %   *  [METRICS] (Dashboard delivery, experience years)

[PROJECTS]  1  % % % %   *  [PROJECT_SKILLS]  *  % % % %   1  [SKILLS]

[EDUCATION] 1  % % % %   *  [ACHIEVEMENTS] (BCA coursework & NCC honors)

[CONTACT_MESSAGES] 1 % % % %   *  [CONTACT_REPLIES] (Email communication trail)
```

7. FLAGSHIP PROJECT SHOWCASE

Detailed breakdown of featured analytical implementations

Sales Performance & Customer Analytics Dashboard (Flagship)

Power BI | Advanced Excel | MySQL | DAX Measures | ETL Pipeline

Conducted end-to-end exploratory and predictive analysis across 50,000+ enterprise sales records representing $2.4M in revenue. Designed dynamic KPI cards, customer acquisition cohorts, profit margin waterfalls, and automated monthly forecasting models with 96.4% precision.

PG Rental Management Web Application

React.js | Node.js | MySQL | RESTful API | Role-Based Access Control

Engineered a full-stack property management platform featuring room allocation matrices, automated monthly billing, tenant verification workflows, and relational schema optimization across tenants, leases, and payments.

Healthcare Patient Inpatient & Financial Risk Analysis

Python | Pandas | NumPy | Matplotlib | Statistical Aggregation

Cleaned and evaluated hospital admission datasets to uncover length-of-stay bottlenecks, insurance claim rejection correlations, and financial risk indicators for clinical operational efficiency.

8. COMPREHENSIVE TECHNOLOGY STACK MATRIX

Modular technologies utilized across the software lifecycle

- **Frontend & UI:** React 18, TypeScript, Vite, Tailwind CSS, Motion (Framer Motion), Three.js WebGL, Lucide Icons
- **Backend & APIs:** Node.js, Express.js, TypeScript, RESTful Endpoints, Server-Sent Events (SSE), Nodemailer TLS SMTP
- **Databases & ORM:** MySQL Relational Database, Connection Pooling, Raw SQL Prepared Statements, Automated Migrations
- **Data Analytics & BI:** Power BI, DAX Measures, Advanced Excel (Pivot/XLOOKUP), Python (Pandas/NumPy), SQL Aggregations
- **DevOps & Hosting:** Vercel (Client Portal), Netlify (Admin Console), Cloud Microservices, Git/GitHub CI/CD
- **Security & Auth:** JWT Token Verification, Bcrypt Password Encryption, Rate Limiting, Input Sanitization